

Project requirements: Students are expected to form groups of 2-3 students and identify an interesting problem, which can benefit from an optimization formulation. The problem can come a wide range of fields, including data science, engineering design, finance, machine learning, medicine or other fields of science. The problem must involve a significant optimization problem that is solved using a numerical method. For instance, you can set up a new machine learning task as an optimization problem with a suitable objective function of your own and solve it using different numerical methods.

Goal: The goal of the project is to provide students with an opportunity to go through three phases involved in solving problems using optimization, including machine learning: (a) Identify an interesting, relevant problem, (b) formulate an appropriate mathematical optimization problem for the identified problem, and (c) solve the optimization problem using a numerical method/solver. By going through these three phases, the students will learn some of challenges that one may encounter in formulating a suitable optimization problem and setting it up for a numerical solver and in solving non-trivial optimization problems using optimization packages.

Deliverables: Each student is required to submit the following:

- i. **Mid-semester report (20 pts, due on April 4, 2026):** The report should include a description of the problem and a motivation for studying the problem. In addition, it should contain a mathematical optimization problem associated with the problem, including the objective function and constraints. The mathematical formulation need not be the final formulation that will be used in the final report, but should demonstrate that the group has given the problem some thought.
- ii. **Final report (60 pts, due on May 9, 2026):** The final report should follow the format of a technical paper (e.g., an IEEE or ACM conference paper or journal article). It should include a clear and detailed description of (a) the problem and motivation for studying it (this could be a refined version of what was included in the mid-semester report), (b) the final mathematical optimization problem with optimization variables and constraints (as well as the reasons for the constraints), (c) detailed numerical results, and (d) a discussion of challenges that the student faced while working on the problem and how the student coped with them and any interesting lessons or observation from the problem. The final report should be at least 3-4 pages (in the aforementioned technical paper format). You can download the templates at <https://www.ieee.org/conferences/publishing/templates>.
- iii. **Oral presentation (20 pts):** Each group is required to give a 7-8 minute presentation on the project during the last lecture (May 5, 2026). The presentation slides should be uploaded to Canvas (a separate assignment has been created for submitting the slides) by May 4, 2026.